HRMS

09 招募管理 — 系統設計書

HR09 RECRUITMENT Service — System Design

系統名稱 **人力資源暨專案管理系統**	版本 **1.0**
日期 **2026-03-18**	作者 **HRMS 開發團隊**

目錄

招募管理服務系統設計書

版本: 1.0 **日期:** 2025-12-07 **Domain代號:** 09 (REC) **導入階段:** 第三階段（進階人資功能）

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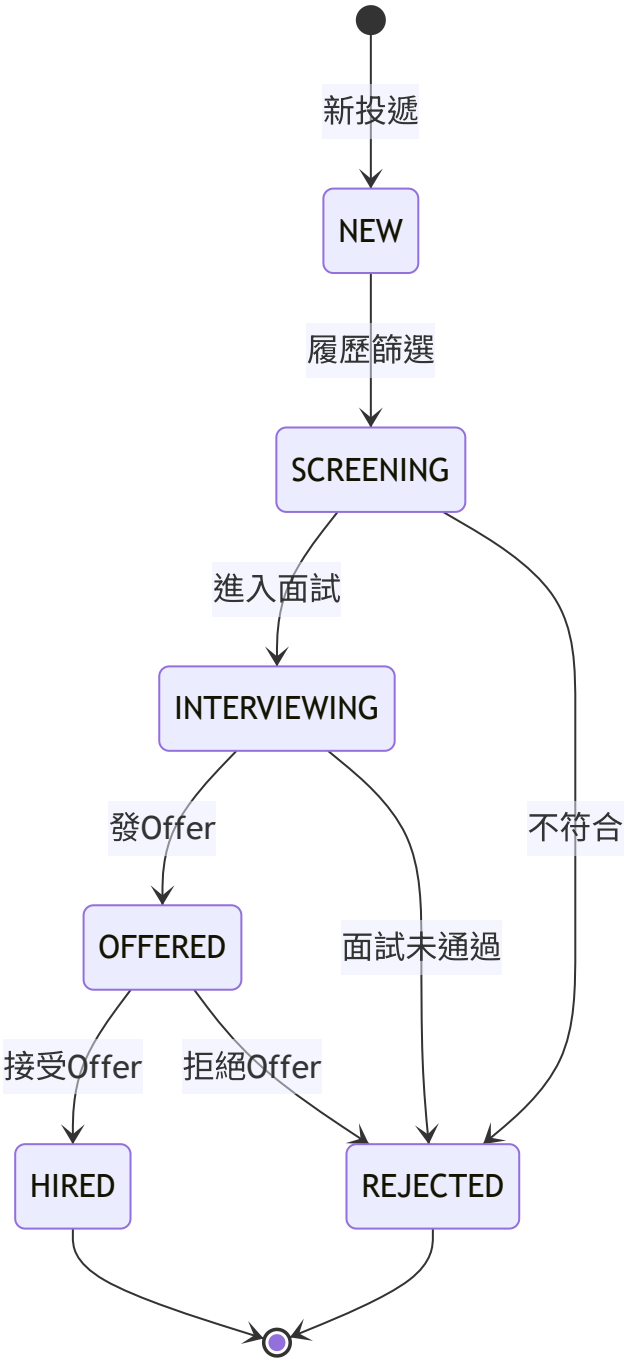
1. 服務概述

1.1 核心功能

- ✅ **職缺需求管理:** 需求提出與審核
- ✅ **應徵者管理:** 履歷、來源、狀態追蹤
- ✅ **面試管理:** 排程、評估、結果彙整
- ✅ **Offer管理:** 發放、接受/拒絕
- ✅ **招募成效分析:** 來源、轉換率

1.2 應徵者狀態流程


新投遞
NEW
履歷篩選
SCREENING
進入面試
INTERVIEWING
發Offer
不符合
OFFERED
面試未通過
接受Offer
拒絕Offer
HIRED
REJECTED

2. UI設計

2.1 頁面清單

頁面代碼	頁面名稱	路由
`HR09-P01`	職缺管理頁面	`/admin/recruitment/jobs`
`HR09-P02`	應徵者管理頁面	`/admin/recruitment/candidates`
`HR09-P03`	應徵者詳情頁面	`/admin/recruitment/candidates/:id`
`HR09-P04`	面試管理頁面	`/admin/recruitment/interviews`
`HR09-P05`	Offer管理頁面	`/admin/recruitment/offers`
`HR09-P06`	招募儀表板	`/admin/recruitment/dashboard`

2.2 UI線稿

2.2.1 應徵者管理頁面 (HR09-P02)

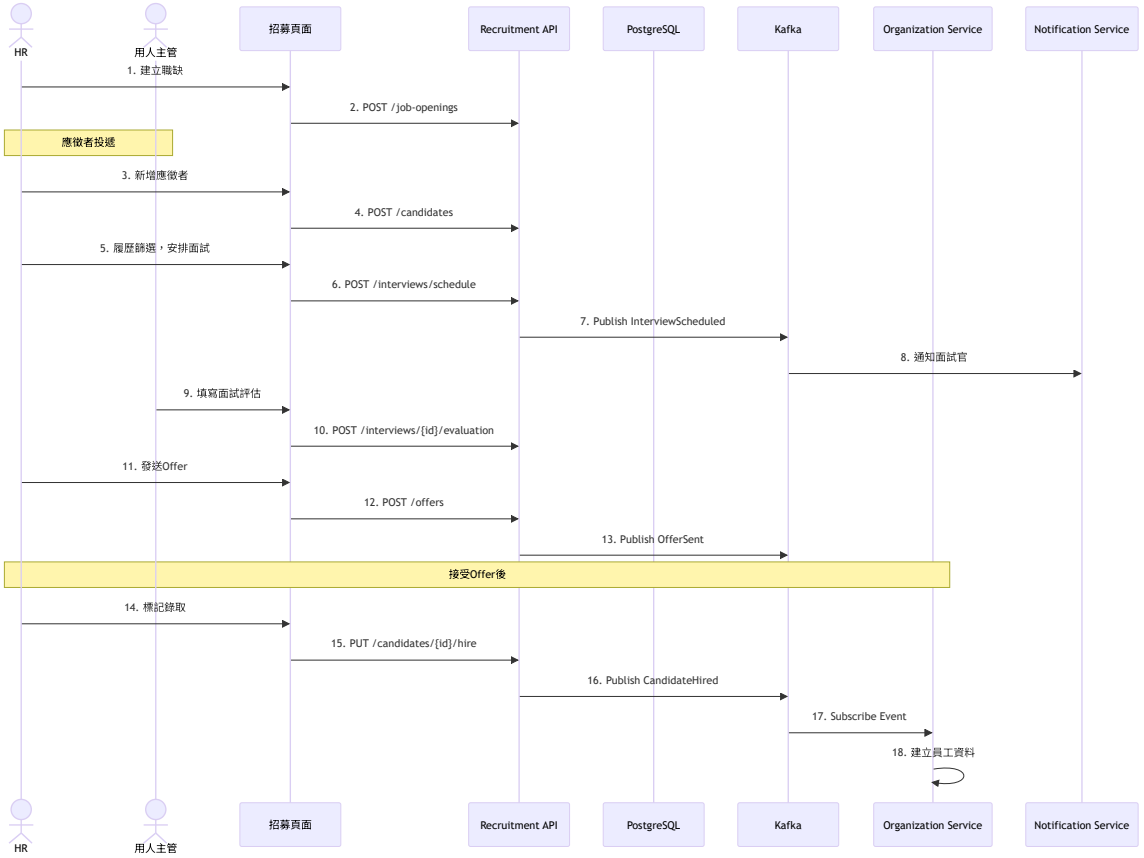
```
┌──────────────────────────────────────────────────────────────┐
│                        應徵者管理頁面                        │
│                                                              │
│                      應徵者管理                              │
│                      ═══════════                             │
│                                                              │
│                          ↓                                   │
│                                                              │
│  🔍 姓名: [_________] | 職缺: [全部▼] | 狀態: [全部▼] | 來源: [全部▼]  │
│                                                              │
│                          ↓                                   │
│                                                              │
│                      看板視圖                                │
│                                                              │
│  ┌──────────┬──────────┬──────────┬──────────┬──────────┐   │
│  │ 新投遞   │ 篩選中   │ 面試中   │ 已發Offer│ 已錄取   │   │
│  │ (15)     │ (8)      │ (5)      │ (2)      │ (1)      │   │
│  ├──────────┼──────────┼──────────┼──────────┼──────────┤   │
│  │┌────────┐│┌────────┐│┌────────┐│┌────────┐│┌────────┐│   │
│  │張三    │王五    │趙六    │孫七    │周八    │   │
│  │前端工程│後端工程│QA工程  │PM      │前端工程│   │
│  │└────────┘│└────────┘│└────────┘│└────────┘│└────────┘│   │
│  │ ┌──────┐ │ ┌──────┐ │ ┌──────┐ │ ┌──────┐ │          │   │
│  │ 李四   │ 陳九   │ 吳十   │ …      │          │   │
│  │ └──────┘ │ └──────┘ │ └──────┘ │ └──────┘ │          │   │
│  └──────────┴──────────┴──────────┴──────────┴──────────┘   │
│                                                              │
│                      拖曳卡片以更新狀態                      │
└──────────────────────────────────────────────────────────────┘
```

2.2.2 招募儀表板 (HR09-P06)

```
┌─────────────────────────────────────────────────────────────────────────────┐
│                                 招募儀表板                                  │
│                                                                             │
│                          ┌──────────────────────────────┐                   │
│                          │        招募成效分析          │                   │
│                          │──────────────────────────────│                   │
│                                          │                                  │
│                                          ▼                                  │
│                          ┌──────────────────────────────────────────────┐   │
│                          │                 關鍵指標                     │   │
│                          │──────────────────────────────────────────────│   │
│                          │ ┌──────────┬──────────┬──────────┬──────────┐ │   │
│                          │ │ 開放職缺 │ 本月投遞 │ 本月面試 │ 本月錄取 │ │   │
│                          │ │    12    │    85    │    23    │    5     │ │   │
│                          │ └──────────┴──────────┴──────────┴──────────┘ │   │
│                                          │                                  │
│                                          ▼                                  │
│                          ┌──────────────────────────────────────────────┐   │
│                          │                 分析圖表                     │   │
│                          │──────────────────────────────────────────────│   │
│                          │ ┌──────────────────────┬──────────────────────┐ │   │
│                          │ │ 📊 來源分析          │ 📈 轉換漏斗          │ │   │
│                          │ │                      │                      │ │   │
│                          │ │ 104人力銀行: 45%     │ 投遞 → 面試: 27%     │ │   │
│                          │ │ 員工推薦: 25%        │ 面試 → Offer: 22%    │ │   │
│                          │ │ 官網: 20%            │ Offer → 錄取: 80%    │ │   │
│                          │ │ LinkedIn: 10%        │                      │ │   │
│                          │ └──────────────────────┴──────────────────────┘ │   │
│                          └──────────────────────────────────────────────┘   │
└─────────────────────────────────────────────────────────────────────────────┘
```

3. UX流程設計

3.1 招募流程



4. 畫面事件說明

4.1 職缺管理頁面事件 (HR09-P01)

事件ID	觸發元素	事件類型	事件處理	後端API
`E-JOB-01`	新增職缺按鈕	onClick	開啟新增職缺對話框	-
`E-JOB-02`	編輯按鈕	onClick	開啟編輯對話框	GET /api/v1/recruitment/job-openings/{id}
`E-JOB-03`	關閉職缺按鈕	onClick	確認後關閉職缺	PUT /api/v1/recruitment/job-openings/{id}/close
`E-JOB-04`	新增職缺確認	onClick	建立職缺	POST /api/v1/recruitment/job-openings
`E-JOB-05`	狀態篩選器	onChange	過濾職缺列表	-

E-JOB-04 詳細流程:

```
const handleCreateJob = async (values: JobOpeningForm) => {
  try {
    await recruitmentService.createJobOpening({
      jobTitle: values.jobTitle,
      departmentId: values.departmentId,
      numberOfPositions: values.numberOfPositions,
      salaryRange: values.salaryRange,
      requirements: values.requirements,
      responsibilities: values.responsibilities
    });

    message.success('職缺已建立');
    await fetchJobOpenings();
    setModalVisible(false);
  } catch (error) {
    message.error('建立失敗: ' + error.message);
  }
};
```

4.2 應徵者管理頁面事件 (HR09-P02)

事件ID	觸發元素	事件類型	事件處理	後端API
`E-CAND-01`	姓名搜尋框	onChange (debounce 300ms)	過濾應徵者	-
`E-CAND-02`	職缺篩選器	onChange	過濾應徵者	GET /api/v1/recruitment/candidates?openingId={id}
`E-CAND-03`	狀態篩選器	onChange	過濾應徵者	-
`E-CAND-04`	來源篩選器	onChange	過濾應徵者	-
`E-CAND-05`	新增應徵者按鈕	onClick	開啟新增對話框	-
`E-CAND-06`	看板卡片拖曳	onDragEnd	更新應徵者狀態	PUT /api/v1/recruitment/candidates/{id}/status
`E-CAND-07`	卡片點擊	onClick	跳轉至應徵者詳情頁	-

E-CAND-06 詳細流程:

```
const handleDragEnd = async (result: DropResult) => {
  const { source, destination, draggableId } = result;

  if (!destination || source.droppableId === destination.droppableId) {
    return;
  }

  try {
    // 更新本地狀態(樂觀更新)
    const newStatus = destination.droppableId as CandidateStatus;
    updateCandidateStatusLocally(draggableId, newStatus);

    // 呼叫API
    await recruitmentService.updateCandidateStatus(draggableId, {
      newStatus,
      reason: `從${source.droppableId}移動至${destination.droppableId}`
    });

    message.success('狀態已更新');
  } catch (error) {
    // 回復狀態
    revertCandidateStatus(draggableId);
    message.error('更新失敗: ' + error.message);
  }
};
```

4.3 應徵者詳情頁面事件 (HR09-P03)

事件ID	觸發元素	事件類型	事件處理	後端API
`E-DETAIL-01`	安排面試按鈕	onClick	開啟面試排程對話框	-
`E-DETAIL-02`	發送Offer按鈕	onClick	開啟Offer對話框	-
`E-DETAIL-03`	拒絕按鈕	onClick	開啟拒絕理由對話框	-
`E-DETAIL-04`	安排面試確認	onClick	建立面試	POST /api/v1/recruitment/interviews/schedule
`E-DETAIL-05`	發送Offer確認	onClick	建立Offer	POST /api/v1/recruitment/offers
`E-DETAIL-06`	拒絕確認	onClick	更新狀態為拒絕	PUT /api/v1/recruitment/candidates/{id}/reject

E-DETAIL-04 詳細流程:

```
const handleScheduleInterview = async (values: InterviewForm) => {
  try {
    await recruitmentService.scheduleInterview({
      candidateId,
      interviewRound: values.round,
      interviewType: values.type,
      interviewDate: values.date,
      location: values.location,
      interviewers: values.interviewers
    });

    message.success('面試已安排，將通知面試官');
    await fetchCandidateDetail();
    setInterviewModalVisible(false);
  } catch (error) {
    message.error('安排失敗: ' + error.message);
  }
};
```

4.4 面試管理頁面事件 (HR09-P04)

事件ID	觸發元素	事件類型	事件處理	後端API
`E-INT-01`	日期篩選器	onChange	過濾面試列表	GET /api/v1/recruitment/interviews?date={date}
`E-INT-02`	面試官篩選器	onChange	過濾面試列表	-
`E-INT-03`	面試卡片點擊	onClick	開啟面試詳情Modal	GET /api/v1/recruitment/interviews/{id}
`E-INT-04`	評估按鈕	onClick	開啟評估表單	-
`E-INT-05`	評估確認	onClick	提交面試評估	POST /api/v1/recruitment/interviews/{id}/evaluations
`E-INT-06`	取消面試按鈕	onClick	取消面試	POST /api/v1/recruitment/interviews/{id}/cancel
`E-INT-07`	重新排程按鈕	onClick	開啟重新排程對話框	-
`E-INT-08`	重新排程確認	onClick	更新面試時間	PUT /api/v1/recruitment/interviews/{id}/reschedule

E-INT-03 詳細流程:

```
const handleViewInterview = async (interviewId: string) => {
  try {
    const detail = await recruitmentService.getInterviewDetail(interviewId);
    setCurrentInterview(detail);
    setDetailModalVisible(true);
  } catch (error) {
    message.error('載入失敗: ' + error.message);
  }
};
```

E-INT-05 詳細流程:

```
const handleSubmitEvaluation = async (values: EvaluationForm) => {
  try {
    await recruitmentService.submitEvaluation(interviewId, {
      interviewerId: currentUser.employeeId,
      technicalScore: values.technicalScore,
      communicationScore: values.communicationScore,
      cultureFitScore: values.cultureFitScore,
      overallRating: values.overallRating,
      comments: values.comments,
      strengths: values.strengths,
      concerns: values.concerns
    });

    message.success('評估已送出');
    await fetchInterviews();
    setEvalModalVisible(false);
  } catch (error) {
    message.error('送出失敗: ' + error.message);
  }
};
```

E-INT-08 詳細流程:

```
const handleReschedule = async (values: RescheduleForm) => {
  try {
    await recruitmentService.rescheduleInterview(interviewId, {
      newInterviewDate: values.newDate,
      newLocation: values.location,
      reason: values.reason
    });

    message.success('已重新排程，將通知相關人員');
    await fetchInterviews();
    setRescheduleModalVisible(false);
  } catch (error) {
    message.error('更新失敗: ' + error.message);
  }
};
```

4.5 Offer 管理頁面事件 (HR09-P05)

事件ID	觸發元素	事件類型	事件處理	後端API
`E-OFR-01`	頁面載入	onMount	載入 Offer 列表	GET /api/v1/recruitment/offers
`E-OFR-02`	Offer 卡片點擊	onClick	開啟 Offer 詳情 Modal	GET /api/v1/recruitment/offers/{id}
`E-OFR-03`	接受 Offer 按鈕	onClick	標記應徵者接受	POST /api/v1/recruitment/offers/{id}/accept
`E-OFR-04`	拒絕 Offer 按鈕	onClick	開啟拒絕理由對話框	-
`E-OFR-05`	拒絕確認	onClick	標記應徵者拒絕	POST /api/v1/recruitment/offers/{id}/reject
`E-OFR-06`	撤回 Offer 按鈕	onClick	公司主動撤回 Offer	POST /api/v1/recruitment/offers/{id}/withdraw
`E-OFR-07`	延長到期日按鈕	onClick	開啟延長日期對話框	-
`E-OFR-08`	延長確認	onClick	更新 Offer 到期日	PUT /api/v1/recruitment/offers/{id}/extend

E-OFR-02 詳細流程:

```
const handleViewOffer = async (offerId: string) => {
  try {
    const detail = await recruitmentService.getOfferDetail(offerId);
    setCurrentOffer(detail);
    setOfferDetailModalVisible(true);
  } catch (error) {
    message.error('載入失敗: ' + error.message);
  }
};
```

E-OFR-06 詳細流程 (撤回 Offer):

```
const handleWithdrawOffer = async (offerId: string) => {
  try {
    await Modal.confirm({
      title: '確認撤回 Offer',
      content: '撤回後將無法恢復，確定要撤回此 Offer 嗎？',
      onOk: async () => {
        await recruitmentService.withdrawOffer(offerId);
        message.success('Offer 已撤回');
        await fetchOffers();
      }
    });
  } catch (error) {
    message.error('撤回失敗: ' + error.message);
  }
};
```

E-OFR-08 詳細流程 (延長到期日):

```
const handleExtendOffer = async (values: ExtendOfferForm) => {
  try {
    await recruitmentService.extendOffer(offerId, {
      newExpiryDate: values.newExpiryDate,
      reason: values.reason
    });

    message.success('Offer 到期日已延長');
    await fetchOffers();
    setExtendModalVisible(false);
  } catch (error) {
    message.error('更新失敗: ' + error.message);
  }
};
```

4.6 招募儀表板事件 (HR09-P06)

事件ID	觸發元素	事件類型	事件處理	後端API
`E-DASH-01`	頁面載入	onMount	載入儀表板資料	GET /api/v1/recruitment/dashboard
`E-DASH-02`	日期範圍篩選	onChange	切換日期範圍並重新載入	GET /api/v1/recruitment/dashboard?from={date}&to={date}
`E-DASH-03`	匯出報表按鈕	onClick	下載Excel報表	GET /api/v1/recruitment/dashboard/export

5. Data Flow設計

5.1 前端狀態管理 (Redux)

5.1.1 State結構

```
interface RecruitmentState {
  // 職缺管理
  jobOpenings: {
    list: JobOpening[];
    currentJob: JobOpening | null;
    filters: {
      status: JobStatus;
    };
    loading: boolean;
  };

  // 應徵者管理
  candidates: {
    kanbanData: {
      NEW: Candidate[];
      SCREENING: Candidate[];
      INTERVIEWING: Candidate[];
      OFFERED: Candidate[];
      HIRED: Candidate[];
      REJECTED: Candidate[];
    };
    currentCandidate: CandidateDetail | null;
    filters: {
      search: string;
      openingId: string;
      status: CandidateStatus;
      source: RecruitmentSource;
    };
    loading: boolean;
  };

  // 面試管理
  interviews: {
    list: Interview[];
    upcoming: Interview[];
    filters: {
      date: string;
      interviewerId: string;
    };
    loading: boolean;
  };

  // Offer管理
  offers: {
```

```
    list: Offer[];
    pending: Offer[];
    loading: boolean;
  };

  // 招募儀表板
  dashboard: {
    kpis: RecruitmentKPIs;
    sourceFunnel: SourceAnalytics[];
    conversionFunnel: ConversionFunnel;
    loading: boolean;
  };
}
```

5.1.2 Redux Actions

```
// 職缺管理Actions
export const jobOpeningActions = {
  fetchJobOpenings: createAsyncThunk(
    'recruitment/fetchJobOpenings',
    async () => {
      const response = await recruitmentService.getJobOpenings();
      return response;
    }
  ),

  createJobOpening: createAsyncThunk(
    'recruitment/createJobOpening',
    async (data: CreateJobOpeningRequest) => {
      const response = await recruitmentService.createJobOpening(data);
      return response;
    }
  ),

  closeJobOpening: createAsyncThunk(
    'recruitment/closeJobOpening',
    async (openingId: string) => {
      await recruitmentService.closeJobOpening(openingId);
      return openingId;
    }
  ),
};

// 應徵者管理Actions
export const candidateActions = {
  fetchCandidates: createAsyncThunk(
    'recruitment/fetchCandidates',
    async (filters: CandidateFilters) => {
      const response = await recruitmentService.getCandidates(filters);
      return response;
    }
  ),

  createCandidate: createAsyncThunk(
    'recruitment/createCandidate',
    async (data: CreateCandidateRequest) => {
      const response = await recruitmentService.createCandidate(data);
      return response;
    }
  ),

  updateCandidateStatus: createAsyncThunk(
    'recruitment/updateCandidateStatus',
    async ({candidateId, status}: {candidateId: string, status: CandidateStatus}) => {
      await recruitmentService.updateCandidateStatus(candidateId, { newStatus: status });
```

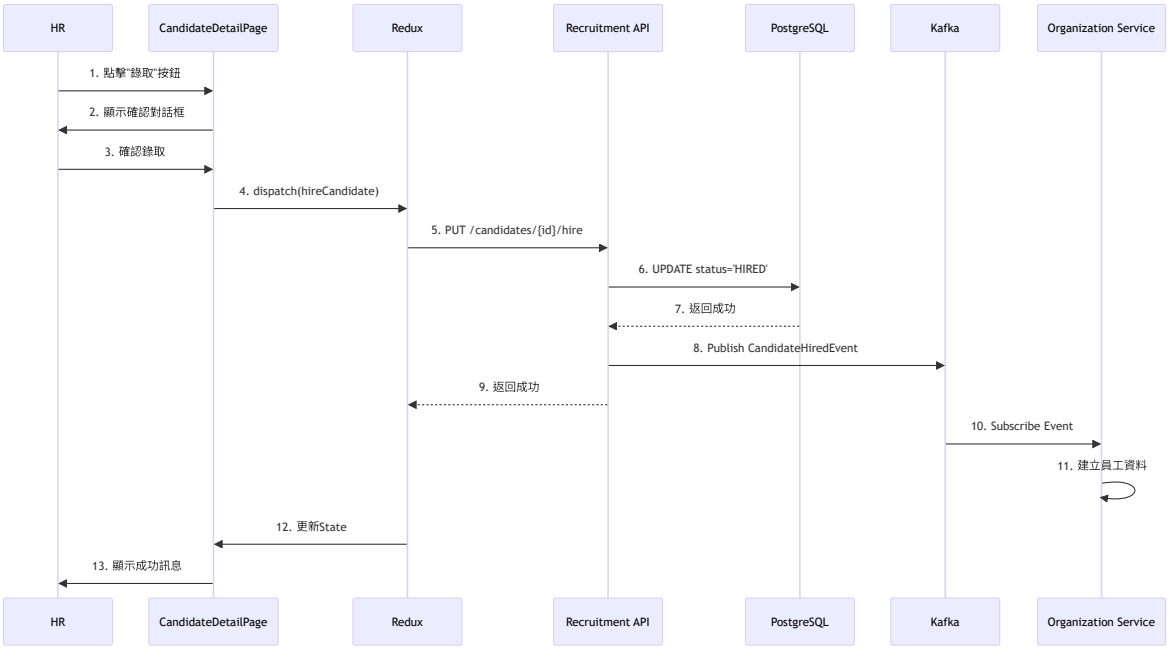
```
      return { candidateId, status };
    }
  ),

  rejectCandidate: createAsyncThunk(
    'recruitment/rejectCandidate',
    async ({candidateId, reason}: {candidateId: string, reason: string}) => {
      await recruitmentService.rejectCandidate(candidateId, reason);
      return candidateId;
    }
  ),

  hireCandidate: createAsyncThunk(
    'recruitment/hireCandidate',
    async (candidateId: string) => {
      await recruitmentService.hireCandidate(candidateId);
      return candidateId;
    }
  ),
};

// 面試管理Actions
export const interviewActions = {
  fetchInterviews: createAsyncThunk(
    'recruitment/fetchInterviews',
    async (filters: InterviewFilters) => {
      const response = await recruitmentService.getInterviews(filters);
      return response;
    }
  ),

  scheduleInterview: createAsyncThunk(
    'recruitment/scheduleInterview',
    async (data: ScheduleInterviewRequest) => {
      const response = await recruitmentService.scheduleInterview(data);
      return response;
    }
  ),

  submitEvaluation: createAsyncThunk(
    'recruitment/submitEvaluation',
    async ({interviewId, data}: {interviewId: string, data: EvaluationRequest}) => {
      await recruitmentService.submitEvaluation(interviewId, data);
      return interviewId;
    }
  ),
};

// Offer管理Actions
export const offerActions = {
  sendOffer: createAsyncThunk(
```

```
    'recruitment/sendOffer',
    async (data: SendOfferRequest) => {
      const response = await recruitmentService.sendOffer(data);
      return response;
    }
  ),

  acceptOffer: createAsyncThunk(
    'recruitment/acceptOffer',
    async (offerId: string) => {
      await recruitmentService.acceptOffer(offerId);
      return offerId;
    }
  ),

  rejectOffer: createAsyncThunk(
    'recruitment/rejectOffer',
    async ({offerId, reason}: {offerId: string, reason: string}) => {
      await recruitmentService.rejectOffer(offerId, reason);
      return offerId;
    }
  ),
};

// 儀表板Actions
export const dashboardActions = {
  fetchDashboard: createAsyncThunk(
    'recruitment/fetchDashboard',
    async (dateRange?: {from: string, to: string}) => {
      const response = await recruitmentService.getDashboard(dateRange);
      return response;
    }
  ),
};
```

5.2 前後端資料流

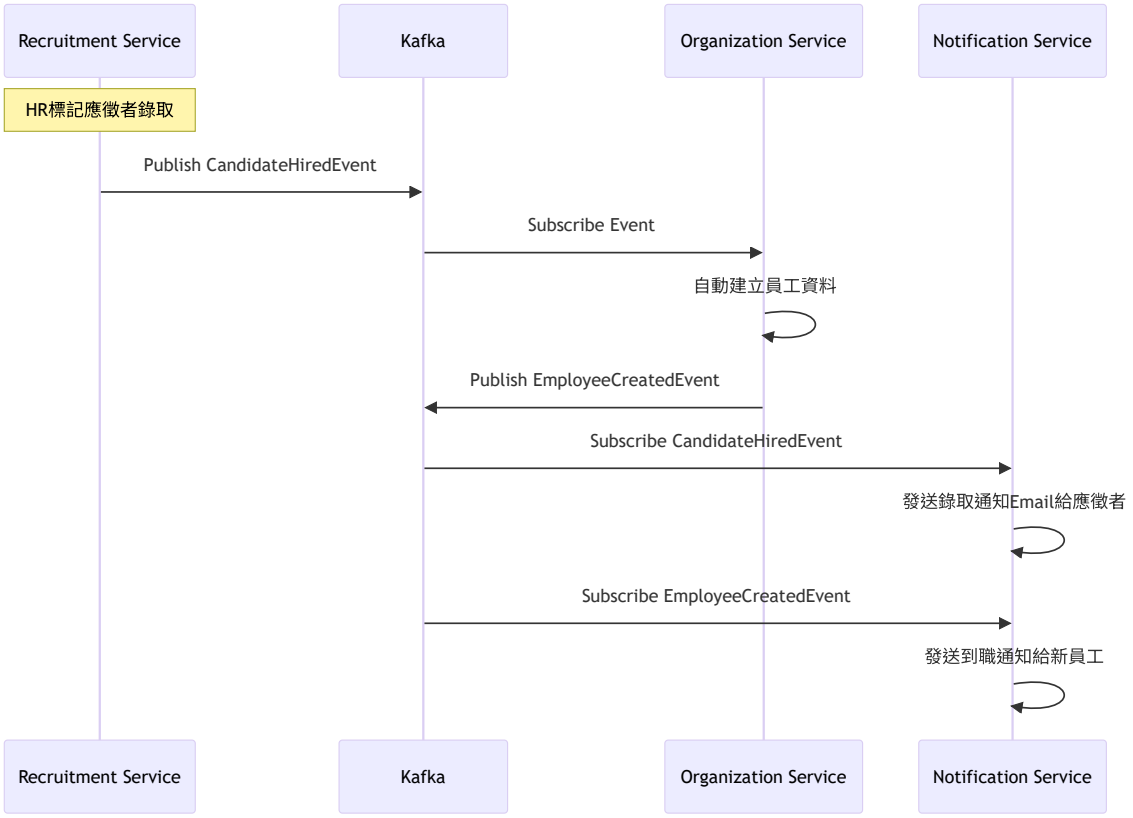
5.2.1 應徵者狀態更新流程 (看板拖曳)

5.2.2 應徵者錄取流程


HR
CandidateDetailPage
Redux
Recruitment API
PostgreSQL
Kafka
Organization Service
1. 點擊"錄取"按鈕
2. 顯示確認對話框
3. 確認錄取
4. dispatch(hireCandidate)
5. PUT /candidates/{id}/hire
6. UPDATE status='HIRED'
7. 返回成功
8. Publish CandidateHiredEvent
9. 返回成功
10. Subscribe Event
11. 建立員工資料
12. 更新State
13. 顯示成功訊息
HR
CandidateDetailPage
Redux
Recruitment API
PostgreSQL
Kafka
Organization Service

5.3 服務間資料流

5.3.1 應徵者錄取事件流



6. 資料庫設計

```
-- 職缺表
CREATE TABLE job_openings (
    opening_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    job_title VARCHAR(100) NOT NULL,
    department_id UUID NOT NULL,
    number_of_positions INTEGER DEFAULT 1,
    salary_range VARCHAR(100),
    requirements TEXT,
    responsibilities TEXT,
    status VARCHAR(20) DEFAULT 'OPEN' CHECK (status IN ('DRAFT', 'OPEN', 'CLOSED', 'FILLED'
    open_date DATE,
    close_date DATE,
    created_by UUID,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 應徵者表
CREATE TABLE candidates (
    candidate_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    opening_id UUID REFERENCES job_openings(opening_id),
    full_name VARCHAR(100) NOT NULL,
    email VARCHAR(255) NOT NULL,
    phone_number VARCHAR(50),
    resume_url VARCHAR(500),
    source VARCHAR(30) CHECK (source IN ('JOB_BANK', 'REFERRAL', 'WEBSITE', 'LINKEDIN', 'OT
    referrer_id UUID,
    application_date DATE DEFAULT CURRENT_DATE,
    status VARCHAR(20) DEFAULT 'NEW'
        CHECK (status IN ('NEW', 'SCREENING', 'INTERVIEWING', 'OFFERED', 'HIRED', 'REJECTED
    rejection_reason TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_candidate_status ON candidates(status);
CREATE INDEX idx_candidate_opening ON candidates(opening_id);

-- 面試表
CREATE TABLE interviews (
    interview_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    candidate_id UUID NOT NULL REFERENCES candidates(candidate_id),
    interview_round INTEGER DEFAULT 1,
    interview_type VARCHAR(30) CHECK (interview_type IN ('PHONE', 'VIDEO', 'ONSITE', 'TECHN
    interview_date TIMESTAMP NOT NULL,
    location VARCHAR(255),
    interviewers JSONB DEFAULT '[]',
    status VARCHAR(20) DEFAULT 'SCHEDULED' CHECK (status IN ('SCHEDULED', 'COMPLETED', 'CAN
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
```

```sql
);

-- 面試評估表
CREATE TABLE interview_evaluations (
    evaluation_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    interview_id UUID NOT NULL REFERENCES interviews(interview_id),
    interviewer_id UUID NOT NULL,
    technical_score INTEGER CHECK (technical_score BETWEEN 1 AND 5),
    communication_score INTEGER CHECK (communication_score BETWEEN 1 AND 5),
    culture_fit_score INTEGER CHECK (culture_fit_score BETWEEN 1 AND 5),
    overall_rating VARCHAR(20) CHECK (overall_rating IN ('STRONG_HIRE', 'HIRE', 'NO_HIRE',
    comments TEXT,
    strengths TEXT,
    concerns TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT uk_evaluation UNIQUE (interview_id, interviewer_id)
);

-- Offer表
CREATE TABLE offers (
    offer_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    candidate_id UUID NOT NULL REFERENCES candidates(candidate_id),
    offered_position VARCHAR(100) NOT NULL,
    offered_salary DECIMAL(12,2) NOT NULL,
    offered_start_date DATE,
    offer_date DATE DEFAULT CURRENT_DATE,
    expiry_date DATE NOT NULL,
    status VARCHAR(20) DEFAULT 'PENDING' CHECK (status IN ('PENDING', 'ACCEPTED', 'REJECTED
    response_date DATE,
    rejection_reason TEXT,
    created_by UUID,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

5. Domain設計

```java
@Entity
public class Candidate {
    @EmbeddedId
    private CandidateId id;

    private UUID openingId;
    private String fullName;
    private String email;
    private String phoneNumber;
    private String resumeUrl;

    @Enumerated(EnumType.STRING)
    private RecruitmentSource source;

    @Enumerated(EnumType.STRING)
    private CandidateStatus status;

    /**
     * 履歷篩選通過
     */
    public void passScreening() {
        if (this.status != CandidateStatus.NEW) {
            throw new DomainException("只有新投遞狀態可以篩選");
        }
        this.status = CandidateStatus.SCREENING;
    }

    /**
     * 進入面試
     */
    public void moveToInterview() {
        this.status = CandidateStatus.INTERVIEWING;
    }

    /**
     * 發送Offer
     */
    public void sendOffer() {
        if (this.status != CandidateStatus.INTERVIEWING) {
            throw new DomainException("只有面試中狀態可以發Offer");
        }
        this.status = CandidateStatus.OFFERED;
    }

    /**
     * 錄取
     */
    public void hire() {
```

```java
        if (this.status != CandidateStatus.OFFERED) {
            throw new DomainException("只有已發Offer狀態可以錄取");
        }
        this.status = CandidateStatus.HIRED;

        DomainEventPublisher.publish(new CandidateHiredEvent(
            this.id.getValue(),
            this.fullName,
            this.email,
            this.phoneNumber
        ));
    }

    /**
     * 拒絕
     */
    public void reject(String reason) {
        this.status = CandidateStatus.REJECTED;
        this.rejectionReason = reason;
    }
}

public enum CandidateStatus {
    NEW, SCREENING, INTERVIEWING, OFFERED, HIRED, REJECTED
}
```

---

8. 領域事件設計

8.1 事件清單

事件名稱	觸發時機	訂閱服務	業務影響
`JobOpeningCreated`	HR建立職缺	-	記錄到招募日誌
`CandidateApplied`	應徵者投遞履歷	-	增加應徵者數量統計
`InterviewScheduled`	HR安排面試	Notification	通知面試官和應徵者
`OfferSent`	HR發送Offer	Notification	通知應徵者收到Offer
`CandidateHired`	HR確認錄取	Organization, Notification	自動建立員工資料、通知相關人員

8.2 事件Schema

8.2.1 JobOpeningCreatedEvent

```
{
  "eventId": "uuid",
  "eventType": "JobOpeningCreated",
  "aggregateId": "opening-id",
  "aggregateType": "JobOpening",
  "occurredAt": "2025-12-07T10:00:00Z",
  "payload": {
    "openingId": "uuid",
    "jobTitle": "前端工程師",
    "departmentId": "dept-uuid",
    "departmentName": "研發部",
    "numberOfPositions": 2,
    "salaryRange": "60K-80K",
    "requirements": "React, TypeScript, 3年以上經驗",
    "openDate": "2025-12-07",
    "createdBy": "hr-user-id",
    "createdByName": "HR Admin"
  },
  "metadata": {
    "userId": "hr-user-id",
    "userName": "HR Admin",
    "source": "Recruitment Service",
    "version": "1.0"
  }
}
```

8.2.2 InterviewScheduledEvent

```
{
  "eventId": "uuid",
  "eventType": "InterviewScheduled",
  "aggregateId": "interview-id",
  "aggregateType": "Interview",
  "occurredAt": "2025-12-07T14:00:00Z",
  "payload": {
    "interviewId": "uuid",
    "candidateId": "candidate-uuid",
    "candidateName": "王小明",
    "candidateEmail": "wang@email.com",
    "openingId": "opening-uuid",
    "jobTitle": "前端工程師",
    "interviewRound": 1,
    "interviewType": "TECHNICAL",
    "interviewDate": "2025-12-15T10:00:00Z",
    "location": "台北辦公室 3F 會議室A",
    "interviewers": [
      {
        "employeeId": "emp-uuid-1",
        "employeeName": "李組長",
        "email": "lee@company.com"
      },
      {
        "employeeId": "emp-uuid-2",
        "employeeName": "張經理",
        "email": "zhang@company.com"
      }
    ],
    "scheduledBy": "hr-user-id"
  },
  "metadata": {
    "userId": "hr-user-id",
    "userName": "HR Admin",
    "source": "Recruitment Service",
    "version": "1.0"
  }
}
```

8.2.3 OfferSentEvent

```
{
  "eventId": "uuid",
  "eventType": "OfferSent",
  "aggregateId": "offer-id",
  "aggregateType": "Offer",
  "occurredAt": "2025-12-07T16:00:00Z",
  "payload": {
    "offerId": "uuid",
    "candidateId": "candidate-uuid",
    "candidateName": "王小明",
    "candidateEmail": "wang@email.com",
    "openingId": "opening-uuid",
    "offeredPosition": "前端工程師",
    "offeredSalary": 70000,
    "offeredStartDate": "2026-01-01",
    "offerDate": "2025-12-07",
    "expiryDate": "2025-12-21",
    "offerDocumentUrl": "https://storage.company.com/offers/xxxxx.pdf"
  },
  "metadata": {
    "userId": "hr-user-id",
    "userName": "HR Admin",
    "source": "Recruitment Service",
    "version": "1.0"
  }
}
```

8.2.4 CandidateHiredEvent (重要事件)

```
{
  "eventId": "uuid",
  "eventType": "CandidateHired",
  "aggregateId": "candidate-id",
  "aggregateType": "Candidate",
  "occurredAt": "2025-12-07T17:00:00Z",
  "payload": {
    "candidateId": "uuid",
    "fullName": "王小明",
    "email": "wang@email.com",
    "phoneNumber": "0912-345-678",
    "openingId": "opening-uuid",
    "jobTitle": "前端工程師",
    "departmentId": "dept-uuid",
    "departmentName": "研發部",
    "offeredSalary": 70000,
    "expectedStartDate": "2026-01-01",
    "resumeUrl": "https://storage.company.com/resumes/xxxxx.pdf",
    "referrerId": "emp-uuid",
    "referrerName": "陳推薦人",
    "hiredBy": "hr-user-id",
    "hiredByName": "HR Admin",
    "hiredAt": "2025-12-07T17:00:00Z"
  },
  "metadata": {
    "userId": "hr-user-id",
    "userName": "HR Admin",
    "source": "Recruitment Service",
    "version": "1.0"
  }
}
```

8.3 事件處理範例

8.3.1 Notification Service 處理 InterviewScheduledEvent

```
@Service
public class RecruitmentEventListener {
    @Autowired
    private NotificationService notificationService;

    @KafkaListener(topics = "recruitment.interview.scheduled")
    public void handleInterviewScheduled(InterviewScheduledEvent event) {
        var payload = event.getPayload();

        // 通知應徵者
        notificationService.send(
            NotificationRequest.builder()
                .recipientEmail(payload.getCandidateEmail())
                .channel(NotificationChannel.EMAIL)
                .template("INTERVIEW_SCHEDULED_CANDIDATE")
                .data(Map.of(
                    "candidateName", payload.getCandidateName(),
                    "jobTitle", payload.getJobTitle(),
                    "interviewDate", payload.getInterviewDate(),
                    "location", payload.getLocation(),
                    "interviewers", payload.getInterviewers()
                ))
                .build()
        );

        // 通知所有面試官
        payload.getInterviewers().forEach(interviewer -> {
            notificationService.send(
                NotificationRequest.builder()
                    .recipientId(interviewer.getEmployeeId())
                    .channel(NotificationChannel.EMAIL)
                    .template("INTERVIEW_SCHEDULED_INTERVIEWER")
                    .data(Map.of(
                        "interviewerName", interviewer.getEmployeeName(),
                        "candidateName", payload.getCandidateName(),
                        "interviewDate", payload.getInterviewDate(),
                        "location", payload.getLocation()
                    ))
                    .build()
            );
        });
    }
}
```

8.3.2 Organization Service 處理 CandidateHiredEvent

```java
@Service
public class CandidateHiredEventListener {
    @Autowired
    private EmployeeService employeeService;

    @KafkaListener(topics = "recruitment.candidate.hired")
    public void handleCandidateHired(CandidateHiredEvent event) {
        var payload = event.getPayload();

        // 自動建立員工資料
        try {
            employeeService.createFromCandidate(
                CreateEmployeeFromCandidateRequest.builder()
                    .fullName(payload.getFullName())
                    .companyEmail(generateCompanyEmail(payload.getFullName()))
                    .personalEmail(payload.getEmail())
                    .mobilePhone(payload.getPhoneNumber())
                    .departmentId(payload.getDepartmentId())
                    .jobTitle(payload.getJobTitle())
                    .hireDate(payload.getExpectedStartDate())
                    .employmentType(EmploymentType.FULL_TIME)
                    .employmentStatus(EmploymentStatus.PROBATION)
                    .probationEndDate(payload.getExpectedStartDate().plusMonths(3))
                    .baseSalary(payload.getOfferedSalary())
                    .referredBy(payload.getReferrerId())
                    .resumeUrl(payload.getResumeUrl())
                    .build()
            );

            log.info("Employee created from candidate: {}", payload.getFullName());
        } catch (Exception e) {
            log.error("Failed to create employee from candidate", e);
            // 發布失敗事件，由HR手動處理
        }
    }

    private String generateCompanyEmail(String fullName) {
        // 簡單實作：姓名拼音 + @company.com
        // 實際需要檢查重複等邏輯
        return fullName.toLowerCase() + "@company.com";
    }
}
```

7. API設計 (14個端點)

端點	方法	Controller
`/api/v1/recruitment/job-openings`	POST	HR09JobCmdController
`/api/v1/recruitment/job-openings`	GET	HR09JobQryController
`/api/v1/recruitment/candidates`	POST	HR09CandidateCmdController
`/api/v1/recruitment/candidates`	GET	HR09CandidateQryController
`/api/v1/recruitment/candidates/{id}`	GET	HR09CandidateQryController
`/api/v1/recruitment/candidates/{id}/status`	PUT	HR09CandidateCmdController
`/api/v1/recruitment/candidates/{id}/hire`	PUT	HR09CandidateCmdController
`/api/v1/recruitment/interviews/schedule`	POST	HR09InterviewCmdController
`/api/v1/recruitment/interviews/{id}/evaluation`	POST	HR09InterviewCmdController
`/api/v1/recruitment/interviews`	GET	HR09InterviewQryController
`/api/v1/recruitment/offers`	POST	HR09OfferCmdController
`/api/v1/recruitment/offers/{id}/accept`	PUT	HR09OfferCmdController
`/api/v1/recruitment/offers/{id}/reject`	PUT	HR09OfferCmdController
`/api/v1/recruitment/dashboard`	GET	HR09ReportQryController

10. 工項清單摘要

10.1 前端開發工項 (總計: 96小時)

工項ID	工項名稱	類別	工時(小時)	負責模組	說明
F-REC-01	**職缺管理頁面 (HR09-P01)**	**頁面開發**	**14**	**features/recruitment**	
F-REC-01-01	JobOpeningListComponent	元件	3	components/	職缺列表表格
F-REC-01-02	CreateJobModal	元件	4	components/	新增職缺對話框
F-REC-01-03	useJobManagement	Hook	3	hooks/	職缺管理邏輯
F-REC-01-04	JobViewModelFactory	Factory	2	factory/	DTO轉換
F-REC-01-05	單元測試	測試	2	tests/	Factory + Component測試
F-REC-02	**應徵者管理頁面 (HR09-P02)**	**頁面開發**	**24**	**features/recruitment**	
F-REC-02-01	KanbanBoardComponent	元件	6	components/	看板視圖(含拖曳)
F-REC-02-02	CandidateCard	元件	3	components/	應徵者卡片
F-REC-02-03	CandidateFilterBar	元件	3	components/	篩選工具列
F-REC-02-04	AddCandidateModal	元件	4	components/	新增應徵者對話框
F-REC-02-05	useCandidateKanban	Hook	4	hooks/	看板拖曳邏輯
F-REC-02-06	CandidateViewModelFactory	Factory	2	factory/	DTO轉換
F-REC-02-07	單元測試	測試	2	tests/	拖曳邏輯測試
F-REC-03	**應徵者詳情頁面 (HR09-P03)**	**頁面開發**	**18**	**features/recruitment**	

工項ID	工項名稱	類別	工時(小時)	負責模組	說明
F-REC-03-01	CandidateDetailComponent	元件	5	components/	應徵者詳情
F-REC-03-02	TimelineComponent	元件	4	components/	時間軸
F-REC-03-03	ScheduleInterviewModal	元件	4	components/	安排面試對話框
F-REC-03-04	SendOfferModal	元件	3	components/	發送Offer對話框
F-REC-03-05	單元測試	測試	2	tests/	元件測試
F-REC-04	**面試管理頁面 (HR09-P04)**	**頁面開發**	**16**	**features/recruitment**	
F-REC-04-01	InterviewListComponent	元件	4	components/	面試列表
F-REC-04-02	InterviewEvaluationModal	元件	5	components/	評估表單
F-REC-04-03	InterviewCalendarView	元件	4	components/	日曆視圖
F-REC-04-04	useInterviewManagement	Hook	2	hooks/	面試管理邏輯
F-REC-04-05	單元測試	測試	1	tests/	Hook測試
F-REC-05	**Offer管理頁面 (HR09-P05)**	**頁面開發**	**12**	**features/recruitment**	
F-REC-05-01	OfferListComponent	元件	4	components/	Offer列表
F-REC-05-02	OfferDetailModal	元件	3	components/	Offer詳情
F-REC-05-03	useOfferManagement	Hook	3	hooks/	Offer管理邏輯
F-REC-05-04	單元測試	測試	2	tests/	元件測試
F-REC-06	**招募儀表板頁面 (HR09-P06)**	**頁面開發**	**10**	**features/recruitment**	
F-REC-06-01	RecruitmentDashboard	元件	3	components/	儀表板
F-REC-06-02	FunnelChart	元件	3	components/	漏斗圖

工項ID	工項名稱	類別	工時(小時)	負責模組	說明
F-REC-06-03	SourceAnalyticsChart	元件	2	components/	來源分析圖
F-REC-06-04	單元測試	測試	2	tests/	圖表測試
F-REC-07	**API & Redux 整合**	**整合**	**2**	**features/recruitment**	
F-REC-07-01	RecruitmentApi	API	1	api/	API呼叫封裝
F-REC-07-02	recruitmentSlice	Redux	1	store/	State管理

前端小計: 96小時

10.2 後端開發工項 (總計: 76小時)

工項ID	工項名稱	類別	工時(小時)	負責模組	說明
B-REC-01	**Domain層開發**	**領域模型**	**24**	**domain**	
B-REC-01-01	JobOpening聚合根	聚合根	3	domain/model	職缺管理
B-REC-01-02	Candidate聚合根	聚合根	6	domain/model	應徵者(含狀態機)
B-REC-01-03	Interview聚合根	聚合根	4	domain/model	面試管理
B-REC-01-04	Offer聚合根	聚合根	3	domain/model	Offer管理
B-REC-01-05	領域事件定義	事件	3	domain/event	5個領域事件
B-REC-01-06	Domain單元測試	測試	5	domain/	100%覆蓋率
B-REC-02	**Repository層開發**	**持久化**	**16**	**infrastructure**	
B-REC-02-01	IJobOpeningRepository	介面	1	domain/repository	Repository介面
B-REC-02-02	JobOpeningRepositoryImpl	實作	2	infrastructure/repository	MyBatis實作
B-REC-02-03	ICandidateRepository	介面	1	domain/repository	Repository介面
B-REC-02-04	CandidateRepositoryImpl	實作	4	infrastructure/repository	含看板查詢
B-REC-02-05	IInterviewRepository	介面	1	domain/repository	Repository介面
B-REC-02-06	InterviewRepositoryImpl	實作	3	infrastructure/repository	含評估查詢
B-REC-02-07	IOfferRepository	介面	1	domain/repository	Repository介面
B-REC-02-08	OfferRepositoryImpl	實作	2	infrastructure/repository	MyBatis實作
B-REC-02-09	Repository測試	測試	1	infrastructure/	整合測試

工項ID	工項名稱	類別	工時(小時)	負責模組	說明
B-REC-03	**Application層開發**	**應用服務**	**20**	**application**	
B-REC-03-01	CreateJobOpeningServiceImpl	Command	2	application/service	建立職缺
B-REC-03-02	CreateCandidateServiceImpl	Command	2	application/service	新增應徵者
B-REC-03-03	UpdateCandidateStatusServiceImpl	Command	3	application/service	更新狀態(驗證)
B-REC-03-04	ScheduleInterviewServiceImpl	Command	2	application/service	安排面試
B-REC-03-05	SubmitEvaluationServiceImpl	Command	2	application/service	提交評估
B-REC-03-06	SendOfferServiceImpl	Command	2	application/service	發送Offer
B-REC-03-07	HireCandidateServiceImpl	Command	3	application/service	錄取(發事件)
B-REC-03-08	GetCandidatesServiceImpl	Query	2	application/service	查詢應徵者
B-REC-03-09	GetDashboardServiceImpl	Query	1	application/service	查詢儀表板
B-REC-03-10	Service測試	測試	1	application/	服務測試
B-REC-04	**Interface層開發**	**API**	**12**	**interface**	
B-REC-04-01	HR09JobCmdController	Controller	2	interface/api	職缺Command
B-REC-04-02	HR09CandidateCmdController	Controller	3	interface/api	應徵者Command
B-REC-04-03	HR09InterviewCmdController	Controller	2	interface/api	面試Command

工項ID	工項名稱	類別	工時(小時)	負責模組	說明
B-REC-04-04	HR09OfferCmdController	Controller	2	interface/api	Offer Command
B-REC-04-05	HR09*QryController	Controller	2	interface/api	Query Controllers
B-REC-04-06	Request/Response DTO	DTO	1	interface/api	14個端點DTO
B-REC-05	**事件處理開發**	**事件驅動**	**4**	**infrastructure**	
B-REC-05-01	EventPublisher整合	發布	2	infrastructure/event	Kafka發布
B-REC-05-02	事件序列化測試	測試	2	infrastructure/	確保Schema正確

後端小計: 76小時

10.3 工項總覽

類別	前端工時	後端工時	小計
頁面開發	94	-	94
API整合	2	-	2
Domain層	-	24	24
Repository層	-	16	16
Application層	-	20	20
Interface層	-	12	12
事件處理	-	4	4
總計	**96**	**76**	**172**

10.4 人力配置建議

- **前端工程師:** 1人，預計2.5週完成 (96小時)
- **後端工程師:** 1人，預計2週完成 (76小時)
- **總開發週期:** 3週 (含測試與整合)

10.5 開發優先順序

1. **Phase 1 (Week 1):**

- 後端: Domain層 + Repository層 (40小時)
- 前端: 職缺管理 + 應徵者看板基礎功能 (38小時)

2. **Phase 2 (Week 2):**
   - 後端: Application層 + Interface層 (32小時)
   - 前端: 應徵者詳情 + 面試管理 (34小時)
3. **Phase 3 (Week 3):**
   - 後端: 事件處理 + 整合測試 (4小時)
   - 前端: Offer管理 + 儀表板 + 整合測試 (24小時)

10.6 技術挑戰與風險

挑戰項目	風險等級	解決方案
看板拖曳功能	中	使用 react-beautiful-dnd 函式庫
狀態轉換驗證	中	Domain層實作狀態機模式
CandidateHired事件整合	高	需與Organization Service協調Schema
履歷檔案上傳	低	使用Document Service API

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